

1	(a)	(i)	horizontal speed constant at 8.2 m s^{-1} vertical component of speed = $8.2 \tan 60^\circ$ = 14.2 m s^{-1}	
		(ii)	$14.22 = 2 \times 9.8 \times h$ vertical distance = 10.3 m	
		(iii)	time of descent = $14.2 / 9.8 = 1.45 \text{ s}$ $x = 1.45 \times 8.2$ = 11.9 m	
	(b)	(i)	smooth path curved and above given path hits ground at more acute angle	
		(ii)	smooth path curved and below given path hits ground at steeper angle	
	(c)		b(ii) will take a longer time as its KE at any point is lower and thus average velocity is lower. Since, vertical distance is same, the time take is lower.	